

Control unit truth table

Inputs: opcode[15:12], funct[3:0]. For R-type, ALUOp = funct[1:0]. x = don't care.

Instruction	RegWrite	ALUSrc	ALUOp	MemWrite	MemToReg	Branch	BNE
add	1	0	00	0	0	0	0
sub	1	0	01	0	0	0	0
sll	1	0	10	0	0	0	0
and	1	0	11	0	0	0	0
lw	1	1	00	0	1	0	0
sw	0	1	00	1	x	0	0
addi	1	1	00	0	0	0	0
beq	0	0	01	0	x	1	0
bne	0	0	01	0	x	0	1
jmp	0	x	xx	0	x	0	0

Signal meanings

RegWrite: write the register file at rt/rd (bits 11:8).

ALUSrc: 0 = ALU input B is R[rd]; 1 = B is sext(imm4).

ALUOp: 00 ADD, 01 SUB, 10 SLL, 11 AND.

MemWrite: write R[rd] to data memory at the ALU result address. No MemRead signal is needed (reads have

MemToReg: 0 = write ALU result back; 1 = write loaded data back.

Branch / BNE / Jump: feed the PCSrc logic (see 'ALU & PC logic').

Jump
0
0
0
0
0
0
0
0
0
0
1

↻ no side effects).